Secure File Distribution System with Time-Limited Access Control

Project Documentation

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1. Introduction

In the digital era, file sharing has become an essential part of communication and data exchange. However, most common file sharing methods do not provide adequate control over how long a file remains accessible or who can access it after sharing.

This project presents a Secure File Distribution System that ensures confidentiality and controlled access to shared files. The system applies encryption techniques along with password protection and a time-based expiry mechanism.

2. Problem Statement

Traditional file sharing systems lack mechanisms to enforce access control after a file has been distributed. Files can be accessed indefinitely without restriction.

This project addresses:

- Unauthorized access to shared files
- Lack of expiry mechanism
- Absence of authentication in simple sharing systems

3. Objectives

- Secure files using encryption
- Implement password-based authentication
- Restrict file access using expiry time
- Prevent unauthorized usage
- Maintain access logs

4. Methodology / Working

Phase 1: Secure File

- Select file
- Set password
- Define expiry time
- Encrypt file
- Store metadata

Phase 2: Open File

- Enter file name

- Enter password

- Verify password and expiry

- Decrypt file if valid

5. Implementation Details

Language: C++

Concepts Used:

- File Handling

- XOR Encryption

- Time Management

- Modular Programming

Modules:

- Encryption Module

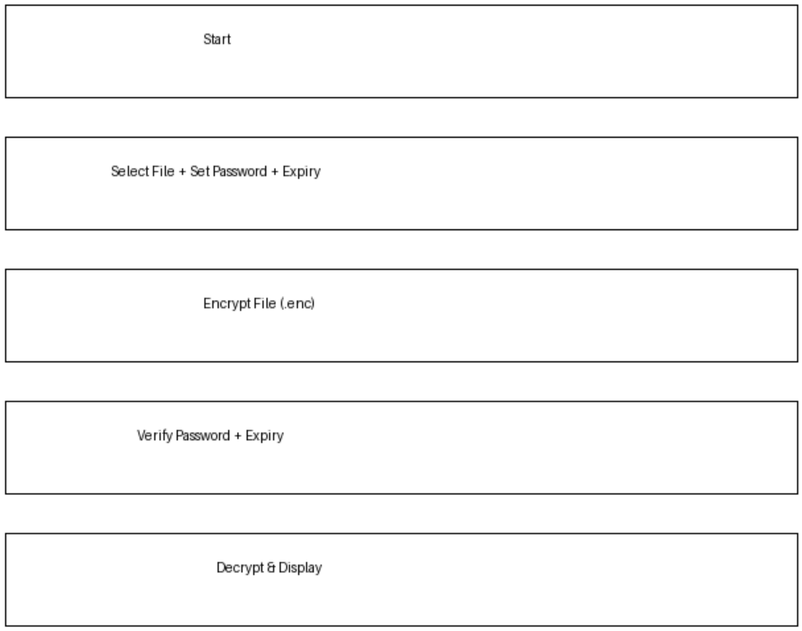
- Access Control Module

- File Manager

- Validator Module

- Logging System

6. System Flowchart



7. Results

The system successfully encrypts and decrypts files. Password verification works correctly and expired files cannot be accessed. Logging tracks all access attempts.

8. Advantages

- Easy to use
- Provides basic security
- Controls file access
- Lightweight system

9. Limitations

- Uses simple XOR encryption
- No network support
- Console-based interface

10. Conclusion

This project demonstrates how encryption and access control can improve file security. It ensures confidentiality and restricts access effectively.

11. Future Enhancements

- Use advanced encryption (AES)
- Add GUI
- Implement cloud sharing
- Improve logging system